

ISC Automated Sales Forecast

AI-powered GM meeting prep — 90 min → under 5 min, every week

v2.6.3-rc1 · Path 1: Built During Challenge

94% Time reduction	90→5 Minutes per week	9 IBM users, live	51 Sessions w/ IBM Bob
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THE PROBLEM

Every week, US Public Sector IBM Sales VP **Dushyant K. Patel** spends 60–90 minutes manually preparing for his GM meeting: opening IBM Sales Cloud (ISC), exporting pipeline data to a spreadsheet, hand-scoring each deal, writing a narrative summary from memory, and formatting a PowerPoint. The process is slow, error-prone, and memory-dependent — and it repeats every single week of every quarter.

IBM Sales Cloud is protected by IBM w3id SSO with passkey authentication, blocking all programmatic access. Seven documented technical approaches — SAQL API, Playwright automation, browser extensions, Salesforce Connected App — all failed. The data was locked inside an authenticated browser session with no extraction path.

THE SOLUTION

ISC Automated Sales Forecast replaces that 90-minute workflow with a single five-minute, AI-powered process. The key insight: a browser HAR file captures authenticated API responses directly — turning IBM's own security posture into a feature. One HAR export, one click, and the entire pipeline is loaded, scored, narrativized, diff'd, and ready to present.

◊ Refresh Data HAR import loads full pipeline in <2 sec. Each user's rows isolated by IBM ID.	📊 Baseline & GM Report One click: immutable baseline written + IBM-branded PPTX downloaded with narrative + diff slide.
○ Score with watsonx ibm/granite-13b scores every deal 0–100. Rule-based fallback when offline.	📄 Custom PPT Column Picker User picks columns, drags to reorder, generates tailored PPTX in seconds.
🔪 GM Narrative llama-3-70b writes meeting-ready opening paragraph + bullets scoped to current filter.	♥ Deal Health Card Hygiene score (0–100), P1/P2/P3 action items, week-over-week timeline per deal.
↕ What Changed Diff engine compares live vs. saved baseline: new deals, stage changes, slippage, amount shifts.	⬆️ Multi-Column Sort Up to 6 stacked sort keys with group-by divider rows. Turns pipeline into analytical workspace.

BUSINESS IMPACT

Client Zero: IBM using its own watsonx technology to transform an internal IBM sales workflow before recommending it to clients. Dushyant Patel's HCLS team within US Public Market IBM is the first production user. The same weekly workflow — forecast review, rep hygiene coaching, GM meeting prep — runs in every IBM geography and every sales segment.

Beyond the primary user: 9 IBM TSLS, ATLS, and the GM/VP are fully provisioned with isolated pipelines. The Pipeline Intelligence Engine surfaces rep coaching signals — blank Next Steps, stale close plans, forecast mismatches — that a VP typically discovers only when it's too late. The permanent quarterly audit ledger creates an SR/SM/Customer trajectory dataset that didn't exist before.

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ARCHITECTURE OVERVIEW

Layer	Technology	Role
Frontend	Vanilla JS · IBM Carbon Design System · IBM Plex Sans	Status-driven action bar, filter bar, modals, drag-to-reorder Column Picker, sort modal, Deal Health Card
Server	Node.js 18 · Express · express-session	All API endpoints, auth middleware, HAR ingestion trigger, PPT generation orchestration
Database	SQLite · better-sqlite3	opportunities, baseline_ledger, users, sessions — all tables keyed by user_id for multi-tenant isolation
Auth	Simulated IBM SSO · bcryptjs · OIDC-ready architecture	Session shape identical to real IBM w3id OIDC — swap to production requires 1 code block + 2 env vars
AI — Scoring	ibm/granite-13b-instruct-v2 via watsonx.ai	0–100 confidence score per deal; rule-based fallback when credentials unavailable
AI — Narrative	meta-llama/llama-3-70b-instruct via watsonx.ai	GM meeting opening paragraph + bullets scoped to current filtered view
AI — Delta	ibm/granite-3-8b-instruct via watsonx.ai	Week-over-week change summary for What Changed slide
PPT	pptxgenjs	IBM-branded PPTX: cover + data slides + What Changed. Dynamic column widths for Column Picker output.
Data Ingestion	HAR parser (custom · scraper/load-from-har.js)	Extracts SAQL API responses from browser HAR — the only viable path past IBM w3id passkey auth

KEY ENGINEERING DECISIONS

HAR ingestion over API scraping: IBM Sales Cloud blocks every programmatic access path. The HAR file captures authenticated browser responses — no credentials stored in the app, no security risk, no IBM IT approval required to get started.

Append-only baseline ledger: Every "Confirm & Generate" writes an immutable UUID-keyed row-set to baseline_ledger. No DELETE or UPDATE paths exist. A strict `conf.i.rmed_at < asOf SQL` guard makes "today vs today" diff structurally impossible — the same discipline as an auditable financial system.

Column Picker PPT: COLUMN_META map drives dynamic proportional width calculation across all 12 fields. `generateCustomPpt()` scales defaultW weights to fill exactly 13.0" — no hardcoded column widths. Uses `filtered.filter(selected)` — the same guard that fixed the 943-vs-21 row bug on the main PPT button.

Pipeline Intelligence Engine: Hygiene scoring (0–100) built from the existing schema — no new data sources. Padded-close detection flags deals within 4 days of quarter-end (P1/P2/P3). Manager-grouped Rep Hygiene Summary with rep drill-down modal. All rules-based; watsonx upgrade path documented in `server/hygieneScore.js`.

Multi-tenant data isolation: Every DB table keyed by `user_id`. PPT output namespaced to `output/(ibm_id)/`. Sessions per-user. Zero cross-user data leakage possible by schema design, not just by access control.

Day 0 reset utility: `scripts/day0-reset.js` wipes all 9 users and re-seeds 54 deals + 9 confirmed baselines in <2 seconds. --dry-run flag previews before writing. Enables reproducible demo and video recording from a known-good state.

BUILD PROCESS

Built entirely with IBM Bob across 51 sessions (~100 hours, July 10–21, 2026). Every design decision, architecture choice, debugging session, and line of production code was collaborative with IBM's own watsonx AI development tool. The complete session log — `UCC1-TechnicalSessionLog.md` — is part of the repository: an unedited technical history of an IBMer and IBM Bob building a production application together.

TEST COVERAGE

34/34 assertions passing — `scripts/test-baseline-ledger.js` runs a full 13-week Q3 2026 in-memory simulation: baseline writes, diff computation, "today vs today" guard, quarter boundaries, multi-user isolation. Passes in under 5 seconds on any Node.js 18+ machine.

Node.js 18	Express	SQLite · better-sqlite3	IBM watsonx.ai	pptxgenjs	IBM Carbon Design System	bcryptjs
express-session	github.com/jefftrbo/UCC1-ISCAutomatedSalesForecast				Tag: v2.6.3-rc1	

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DEVELOPMENT TIMELINE — JULY 10–21, 2026

Jul 10	Kick-off session. HAR ingestion PoC. SQLite schema. Rule-based scoring. First PPTX generated. (Sessions 1–3)
Jul 11–13	IBM Carbon Design System UI. Action bar. Named view presets. Diff engine. Permanent audit ledger. 34/34 test harness. (v2.2.0–v2.3.0)
Jul 14–15	Simulated IBM SSO. Multi-tenant data isolation. 9 test users. Per-user seed data. Login page. Header user chip. (v2.4.0)
Jul 16–18	Pipeline Intelligence Engine. Deal Health Card. Hygiene scoring. Padded-close detection. Manager-grouped Rep Hygiene. Rep drill-down modal. 21 micro-versions. (v2.5.0–v2.5.21)
Jul 18–19	PTMP slide generator. Font polish. Multi-column sort modal with group-by dividers. Narrative fetch guard hotfix. RC2 tagged. (v2.6.0–v2.6.2-rc2)
Jul 21	Column Picker Custom PPT. Two-panel modal, drag-to-reorder, 12 columns, dynamic widths. Day 0 reset utility. Full doc sweep. RC1 tagged. (v2.6.3-rc1)

VERSION SUMMARY

Version	What shipped
v1.0.0 – v2.1.x	HAR ingestion · SQLite · rule-based scoring · first PPTX
v2.2.0 – v2.3.0	IBM Carbon UI · diff engine · permanent audit ledger · 34/34 tests
v2.4.0	Simulated IBM SSO · multi-tenant isolation · 9 users · login page
v2.5.0 – v2.5.21	Pipeline Intelligence Engine · Deal Health Card · Rep Hygiene coaching dashboard
v2.6.0 – v2.6.1	PTMP slide generator (Frank Attaie format) · font + alignment polish
v2.6.2 / rc2	Multi-column sort modal · group-by dividers · narrative hotfix · submission RC
v2.6.3 / rc1	Column Picker Custom PPT · Day 0 reset utility · full doc sweep · final RC

TEAM

Jeffrey L. Trbovich — Technical Lead
Built entire application across 51 sessions (~100 hours) using IBM Bob. IBM ATL / US Public Sector.

Dushyant K. Patel — Business Lead, Sales VP
US Public Market IBM. Business owner, primary user, domain expertise. Live pipeline data used throughout development.

IBM Bob — Co-developer
IBM's watsonx-powered AI development assistant. Co-developer for all 51 sessions. Every line of production code and every architecture decision.

John Albertson · Kim Salatino · Justin Griffin · Kerry Hovan · Spencer Korn · Jeff Underwood · Muhammad Safwat — Challenge team members and test users.

PATH TO PRODUCTION

The **CIO "Build with watsonx" Path to Production** has been identified as the official deployment channel. A ServiceNow AI System Demand is being submitted with the architecture diagram. Upon Technical Review Board approval: IBM Cloud Code Engine deployment, CIO-managed watsonx.ai credentials, real IBM w3id SSO — available to any IBM Sales VP from any browser, no local installation required.

Approval as an official IBM AI System creates standing to request a **Salesforce Connected App with OAuth 2.0** — eliminating the HAR step entirely. The same application, unchanged, serves any IBM Sales VP with a Salesforce-backed pipeline. IBM Public Market is the immediate target; broader IBM Sales organization adoption is the medium-term opportunity.

Repository: `github.com/jefftrbo/UCC1-ISCAutomatedSalesForecast` · Tag: v2.6.3-rc1 · 75+ commits · 51 sessions · ~100 hours · July 10–21, 2026
Test harness: `node scripts/test-baseline-ledger.js` — 34/34 assertions, <5 seconds · **Reset:** `node scripts/day0-reset.js`